

Chirag Aparadh



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Education

University of Massachusetts Amherst

Expected Graduation: May 2028

Master's in Computer Science (M.S.)

Amherst, MA

Ramrao Adik Institute of Technology

2023

Bachelor's in Computer Science & Engineering (B.Tech, CGPA: 9.51 / 10)

Mumbai University

Professional Experience

Oracle Financial Services Software

Jul 2023 – Aug 2026

Software Engineer (Associate Consultant)

Mumbai, MH

- Built and maintained core Java EE microservices powering HDFC Bank's net banking platform (2M+ daily active users), delivering 99.99% uptime through highly available clustered deployments.
- Implemented a BioCatch bot-detection workflow for merchant transactions (e.g., Amazon, Flipkart), triggering OTP authentication on suspicious activity, strengthening fraud prevention for 150K daily transactions.
- Engineered event-driven integrations with Apache Kafka to publish acknowledgment messages to external systems (e.g., Zerodha) after fund transfers, enabling reliable, decoupled communication across banking services.
- Modernized a legacy banking workflow by translating UI-driven business logic into backend service APIs, performing end-to-end debugging across service and database layers and improving reliability through standardized data contracts and validation.

Projects

JiraRL | Python, FastAPI, OpenEnv, Qwen3, QLoRA, GRPO, Hugging Face

Jun 2026 – Aug 2026

- Created a stateful, OpenEnv-compatible Jira simulator for training LLM agents, with structured actions, isolated episodes, dependency constraints, and executable reward functions.
- Generated and validated 3,500 hint-free procedural decisions across disjoint train/val/test seeds, publishing reproducible datasets and adapters to Hugging Face.
- Fine-tuned Qwen3-0.6B with 4-bit QLoRA, completing 60/60 held-out in-distribution episodes with 500/500 productive transitions and zero invalid actions.
- Built environment-backed GRPO with checkpoint recovery and promotion gates; diagnosed v1 action collapse and v2 zero-gradient saturation, blocking an unjustified long run.

Multimodal Image Search System | FastAPI, PyTorch, CLIP, Qdrant, Redis, AWS

Jan 2025 – May 2025

- Developed a FastAPI-based asynchronous image processing service generating 512-dimensional CLIP embeddings for semantic image understanding.
- Indexed image vectors in Qdrant to support text-to-image retrieval, visual similarity search, and near-duplicate detection.
- Designed a distributed image processing pipeline using Celery and Redis for async task execution and real-time status tracking, deployed via Docker, AWS ECR, ECS, and S3.

Technical Skills

Languages: Java, Python, C, JavaScript, SQL

Backend & Cloud: Java EE, FastAPI, Django, React, REST APIs, Microservices, Apache Kafka, Redis, Qdrant, Oracle Database, MySQL, MongoDB, AWS, Docker, Kubernetes, Jenkins, Linux

AI & Core CS: PyTorch, CLIP, Qwen3, QLoRA, GRPO, NumPy, Pandas, Scikit-learn, Hugging Face, OpenEnv, DSA, OOP, Operating Systems, Computer Networks, Distributed Systems

Achievements

- Reached the national finale of the Meta × Hugging Face OpenEnv Hackathon (31,000+ teams).
- Ranked 1108 globally in LeetCode Biweekly Contest 145 and earned the Knight badge (rating: 1866).
- Achieved 5-Star Gold Badge in Problem Solving on HackerRank.
- Cleared GATE CS 2023 with an AIR 2816 among 75,680 candidates.
- Published "Exploring the Role of Blockchain in Enhancing Supply Chain" (IEEE, Sep '23) – [link](#).